

MATH 1190 & 1210 Survey of Calculus I

Lesson 11: Local Extrema & the 1st Derivative Test

Name: Key

Learning Target(s): A1, A2

In this activity, we establish a means of finding local extreme values (also called relative extreme values) of a function using properties of the derivative. First, review some key ideas from the online pre-class activity to prepare yourself for the important ideas of the lesson.

Warm-up exercise 1. Fill in the blanks to complete the definition of a critical number, then respond to the prompt below.

Definition: Critical Number

A **critical number** of a function f is a number c at which f is continuous and either

$$f'(c) = \underline{0} \quad \text{or} \quad f'(c) \text{ does not exist .}$$

TRUE or FALSE? $f(x) = \frac{1}{x}$ has exactly 1 critical number.

(circle one):

TRUE

FALSE

$$f(x) = \frac{1}{x} \Rightarrow f'(x) = -\frac{1}{x^2} = -\frac{1}{x^2} \Rightarrow f'(x) \text{ DNE when } x=0 \Rightarrow \text{BUT } f(x) = \frac{1}{x} \text{ is not continuous at } x=0!$$

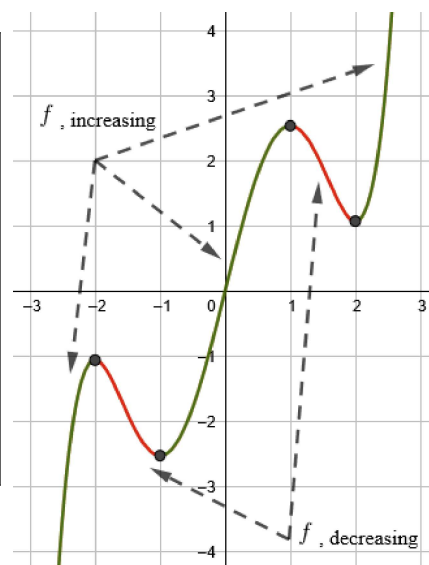
$\& f'(x)=0 \text{ for no } x$

Warm-up exercise 2. Fill in the blanks to complete the statement of the First Derivative Test as it appeared in the online pre-class activity.

First Derivative Test

Suppose that c is a critical number of a continuous function f .

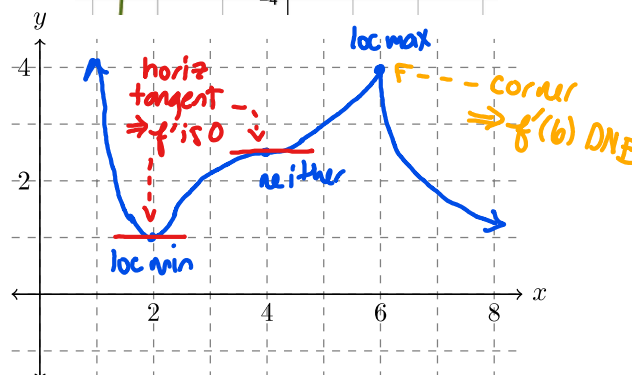
- If f' changes from + to - at c , then f has a local maximum at c .
- If f' changes from - to + at c , then f has a local minimum at c .
- If f' does not change sign at c , then f has neither a local maximum nor local minimum at c .



A1

Warm-up exercise 3. Sketch the graph of a function f satisfying the following:

- $f'(2) = 0$ and $f(2)$ is a local/relative minimum
- $f'(4) = 0$ and $f(4)$ is neither a maximum nor a minimum
- $f'(6)$ DNE and $f(6)$ is a local/relative maximum



Example 1. Let $g(x) = x - \ln(x^2 + 1)$.

(a) Find the critical numbers of g .

$$g'(x) = 1 - \frac{1}{x^2+1} \cdot 2x = 1 - \frac{2x}{x^2+1}$$

$g'(x)$ DNE? $x^2+1=0 \Rightarrow x^2=-1 \Rightarrow$ no real number produces a negative value when squared

$$g'(x)=0? \left(1 - \frac{2x}{x^2+1} = 0\right) \cdot (x^2+1) \Rightarrow 1 \cdot (x^2+1) - \frac{2x}{x^2+1} \cdot (x^2+1) = 0 \cdot (x^2+1) = 0$$

$$\Rightarrow x^2+1-2x=0 \Rightarrow x^2-2x+1=0 \Rightarrow (x-1)^2=0 \Rightarrow x-1=0 \Rightarrow \boxed{x=1}$$

$g(1) = 1 - \ln(2)$ is defined so $\uparrow$ is a crit #!

(b) Find the intervals where g is increasing or decreasing.

$$g'(x) = 1 - \frac{2x}{x^2+1}$$

$g'(x)$ DNE for no real x & $g'(x)=0$ when $x=1$

$\Rightarrow x=1$ is the only value at which g' could change signs!

let's make a sign line to record the sign of g'



$$g'(0) = 1 - \frac{2 \cdot 0}{0^2+1} = 1 - 0 = 1 \oplus \quad \& \quad g'(2) = 1 - \frac{2 \cdot 2}{2^2+1} = 1 - \frac{4}{5} = \frac{5}{5} - \frac{4}{5} = \frac{1}{5} \oplus$$

$\Rightarrow$
 g is increasing on $(-\infty, 1)$ & $(1, \infty)$ (b/c that's where g' is $+$)
 g is never decreasing (b/c g' is never $-$)

(c) At which critical number(s) does g attain a local extreme value(s)? Explicitly cite a relevant test in your response.

Because the sign of g' doesn't change at the critical number $x=1$, g has neither a local max nor local min at $x=1$.

g attains no local extrema.